

PAUL WANG, MECHANICAL ENGINEERING STUDENT

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WORK EXPERIENCE

Technical Director

Edinburgh University Formula Student (EUFS)

JUL 2026 – PRESENT

- Led a 150-member cross-disciplinary team across four departments (Mechanical, Electrical, Software, and Business). Ultimately responsible for the design and manufacture of an autonomous race car (Formula Student ADS), a Concept Class car, and a Dynamic Driving Task (DDT) entry.
- Managed coordination between 18 subteams to work toward achieving the goal of winning the in-house manufactured, fully autonomous car (ADS) division at the Formula Student UK competition.

Controls Engineering Intern

Horwitz LLC, Building Automation Systems

MAY 2026 – SEP 2026

- Industrial air and fluid handling and corporate HVAC controls. Field and office experience.
- Niagara 4 Java custom-built PID controllers and VAV Lynxspring edge BAJA modules.
- Platform and Station network via BACnet MSTP. Java trust store Auth Cert signing and verification.
- AutoCAD component handling and formatting, created custom controller blocks.
- SQL & PowerApps web development. Bespoke dashboard developed as a replacement for Coins ERP.
- Sales Engineering: Excel constrained optimisation algorithm to minimise cost for controller I/O config.

Vehicle Dynamics Engineer

Edinburgh University Formula Student (EUFS)

DEC 2024 – JUL 2026

- Designed a suspension geometry optimiser from scratch using Simscape multibodies. Created full-car kinematic model, determined optimal suspension geometry and roll axis. Validated quantitative performance benefits of anti-ackermann and anti-dive and used in-house tyre magic formula model.
- Design, FEA, & topology optimisation for suspension and wheel components in SolidWorks.
- Acted as Suspension Lead at Formula Student UK design presentation, contributed to team's 6th place, topology optimisation and overall design contributed 15 points and mentioned as good design.
- Manufacturing experience, additive manufacturing, CNC, laser cutting, and hydraulic systems.
- Pioneered emergency braking system hydraulic concepts, including actuation and fluid recovery.

Motorboat, Sailboat, and Aircraft Maintenance Technician

Minneapolis Sailing Center (MSC) & Twin Cities Flight Training (TCFT)

JUL 2023 – PRESENT

- Motorboat & sailboat maintenance at MSC, performed hydraulic steering, drivetrain, and electrical maintenance on motorboats, and fibreglass & carbon fibre composites work for our fleet of sailboats.
- Aircraft repairs, teardowns, and component analysis at TCFT, performed repairs on hydraulics, tyres, brakes, avionics, and fuselage. Performed teardowns on Cessna airframes and engines.

Sailing Instructor

Minneapolis Sailing Center (MSC)

MAY 2022 – PRESENT

- Head coach for the Optimist Racing Team, managed logistics and coached a team of 8-12 students. Solely responsible for designing all lesson plans and tailoring my coaching for each student. Coached Optimist and C420 racing teams at official US Sailing regattas.
- Head instructor for the Adult Laser Racing class, taught complex physics and aeronautical concepts, such as wing-tip vortices, airflow separation, and centre of gravity & roll center dynamics.
- 1500+ work hours, 75+ class sessions taught, named best instructor of 2024.



EDUCATION

BEng Honours Mechanical Engineering

The University of Edinburgh

SEP 2024 – PRESENT

- Grade: First Class (A2). Completed 10 classes with 160 UK credits.

University Classes Taken for High School

University of Minnesota – Twin Cities

SEP 2021 – MAY 2024

- 3.66 Cumulative GPA, 4.0 Technical GPA, 3 Dean's List Honours.
- Completed 17 university classes with 70 US credits. Normal university classes taken full time as secondary school credit. Followed the UMN Mechanical Engineering & Computer Science degree tracks.

Pilot Training

Twin Cities Flight Training (TCFT)

MAY 2022 – JUN 2024

- FAA Licensed Private Pilot. 270 landings, 200 flight hours. Instrument & ILS landings. 30,000+ miles flown.

PROJECTS

Software and Programming Languages

- **MATLAB** Differential equations & matrices calculations for Formula Student.
- **Simulink & Simscape** Simscape multibodies and Simulink for suspension geometry optimiser.
- **SolidWorks** Design, FEA, & topology optimisation for Formula Student.
- **Java** Created Chess, Minesweeper, Tetris, and hashing tables from scratch.
- **Python** Created orbital motion simulations and real-time data processors.
- **HTML & CSS** Used VS-Code to create my personal website: <https://paulhanwang.co.uk>.
- **Niagara 4** BAS controller logic for Lynxspring VAV and JCI JACE. BAJA modules and cert auth.
- **SQL & Power FX** ERP dashboard created in Microsoft PowerApps. Web and mobile app deployment.

Cinematographer & Graphic Designer

- Created several amateur cinematography films and sailboat racing action videos.
- Created bespoke logo, business card, and website design.
- Used Unreal Engine 5 renders for official Edinburgh University Formula Student marketing posters.

Community Outreach Volunteer

- Volunteered at **MSC** as a skipper to give free sailboat rides to members of the public during our Free Sailing Sunday events and local Minneapolis charities during community outreach events.

Academic Outreach Volunteer

- Volunteered for **EUFIS** at Blackhall Primary School, giving a presentation on vehicle dynamics and formula student, and helping the class of P6 students to design and 3D print their own racecars.

LICENCES

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| – Private Pilot Airplane |  Federal Aviation Administration |
| – Motorsport Competition |  Fédération Internationale de l'Automobile |
| – CPR-First Aid (Adult and Pediatric) |  American National Red Cross |
| – Lake Service Provider Watercraft Operation |  Minnesota Department of Natural Resources |
| – OSHA-10 |  United States Department of Labor |